

A PROPOSAL TO DEVELOP SPMVV STUDENT SATELLITE - MECHANICAL ENGINEERING DESIGN ASPECTS

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. ABSTRACT:

The mechanical structure of a student-developed CubeSat serves as the foundational framework, ensuring structural integrity, compatibility with launch dispensers, and protection against the harsh space environment. Standardized under the CubeSat Design Specification (CDS), a basic 1U CubeSat measures 10 cm × 10 cm × 11.35 cm with a mass limit of up to 2 kg (scalable to larger sizes like 3U or 6U for more complex missions). The structure typically consists of anodized aluminium rails (e.g., 6061-T6 alloy for durability and low outgassing) along the corners for sliding into dispensers like the Poly- Picosatellite Orbital Deployer (P-POD), with standoffs at least 6.5 mm high to accommodate switches and protrusions. Key considerations include venting analysis to prevent pressure buildup during ascent, center of gravity within 2 cm of the geometric center, and withstanding loads up to 1200 N in compression. Deployables such as solar panels or antennas are constrained using mechanisms like nichrome burn wires or Hold & Release Mechanisms (HRM) to avoid premature activation during vibration testing.

Materials must comply with orbital debris mitigation (e.g., low-melting-point alloys for reentry burn-up), and designs often incorporate modularity for easy assembly by students, allowing internal mounting of components with flexibility for repairs. Testing involves vibration, shock, thermal vacuum, and static load verifications to meet NASA CSLI and ICD requirements. Student projects emphasize in-house fabrication to reduce costs and enhance learning, with

examples like the UPSat 2U CubeSat using composite materials for lightweight scalability. Innovations include 3D-printed components (e.g., using ABS or PLA polymers in MakerSat missions) to test material resilience in space, demonstrating minimal mass loss under radiation and thermal cycling.